

Week 2 Weekly Report

1. Work Completed

During Week 2, the following tasks were completed:

Cleaned and standardized the collected financial market datasets.

Processed missing values using interpolation methods.

Detected outliers using the IQR method.

Aligned multiple financial time series based on trading dates.

Constructed quantitative features, including daily returns, rolling volatility, moving averages, price momentum, interest rate momentum, and rolling correlations.

Developed reusable preprocessing functions for data cleaning, time series alignment, outlier detection, and feature engineering.

Implemented an automated preprocessing pipeline to support reproducible data processing.

Exported the processed dataset and summary reports for subsequent model implementation.

2. Deliverables

Completed deliverables include:

Week 2 Data Preprocessing Notebook

Automated Preprocessing Pipeline (preprocess.py) and scheduled GitHub Actions workflow

Processed Dataset

Outlier Summary Reports

Week 2 Learning Report

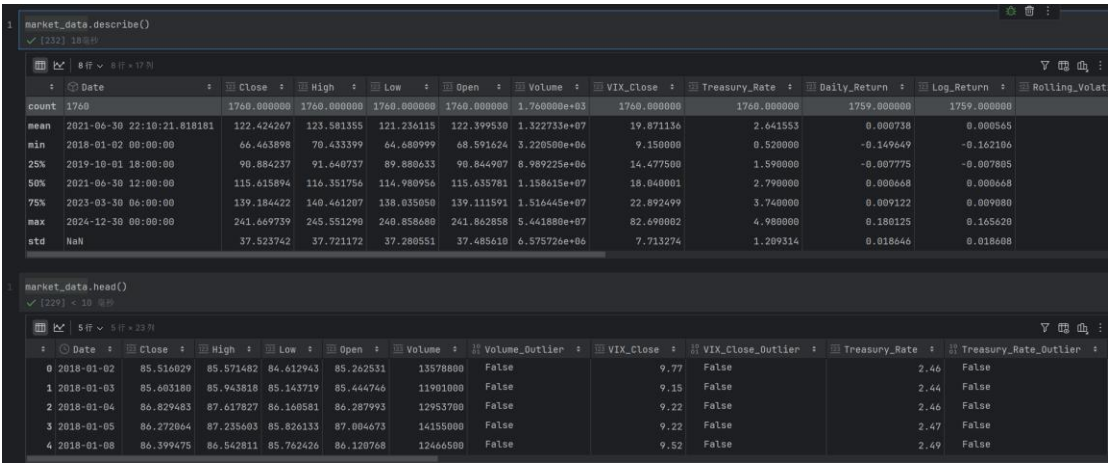
3. Progress Summary

Core preprocessing is complete. A scheduled GitHub Actions workflow now rebuilds and validates the fixed 2018-2024 dataset every Monday and supports manual runs. Historical news sentiment remains the only documented Week 2 data gap.

A complete data preprocessing pipeline has been established for the project. The collected market datasets have been cleaned, standardized, and enriched through feature engineering, providing a reliable dataset for subsequent option pricing model implementation and validation.

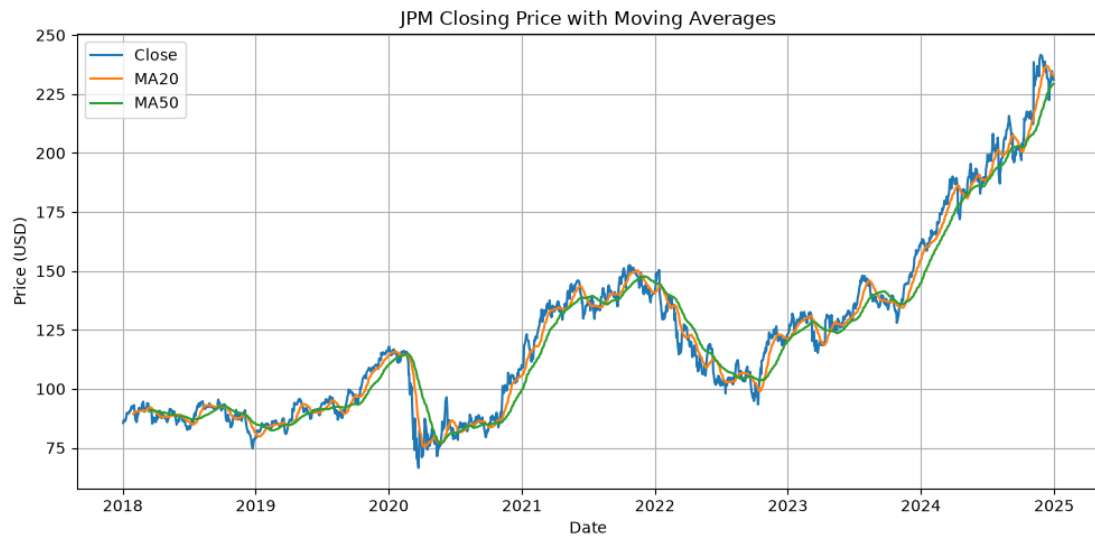
4. Experimental Results

Table 1. Preview of the Processed Dataset



The processed dataset contains cleaned market data together with engineered financial features, providing the input for subsequent option pricing models.

Figure 1. JPM Closing Price (2018–2024)



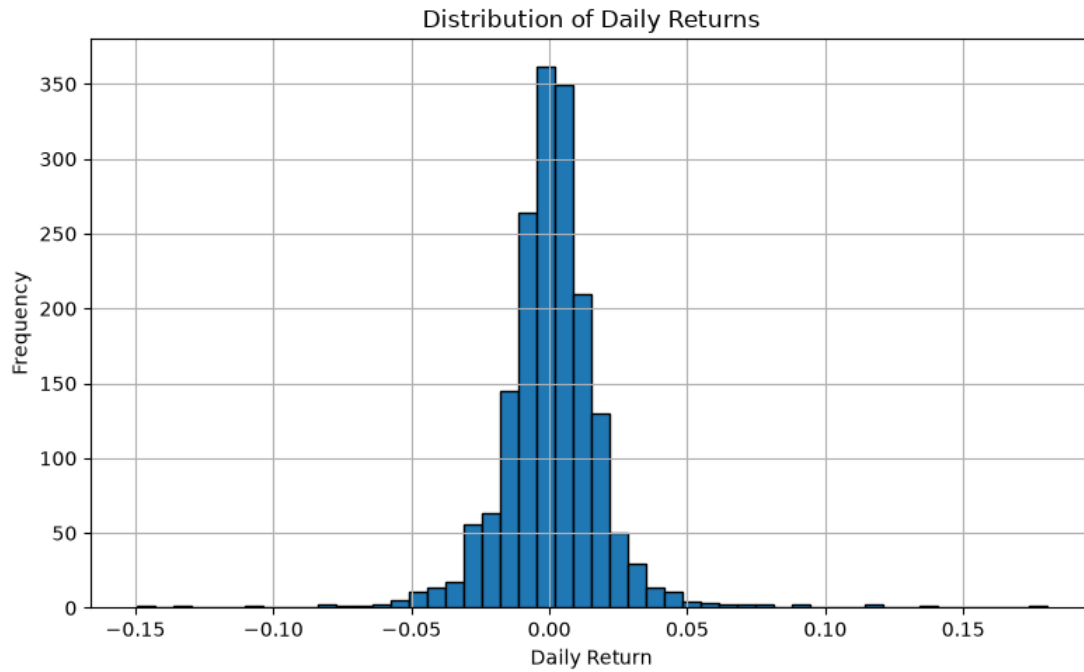
The historical closing prices of JPM after data preprocessing.

Figure 2. JPM Closing Price with Moving Averages



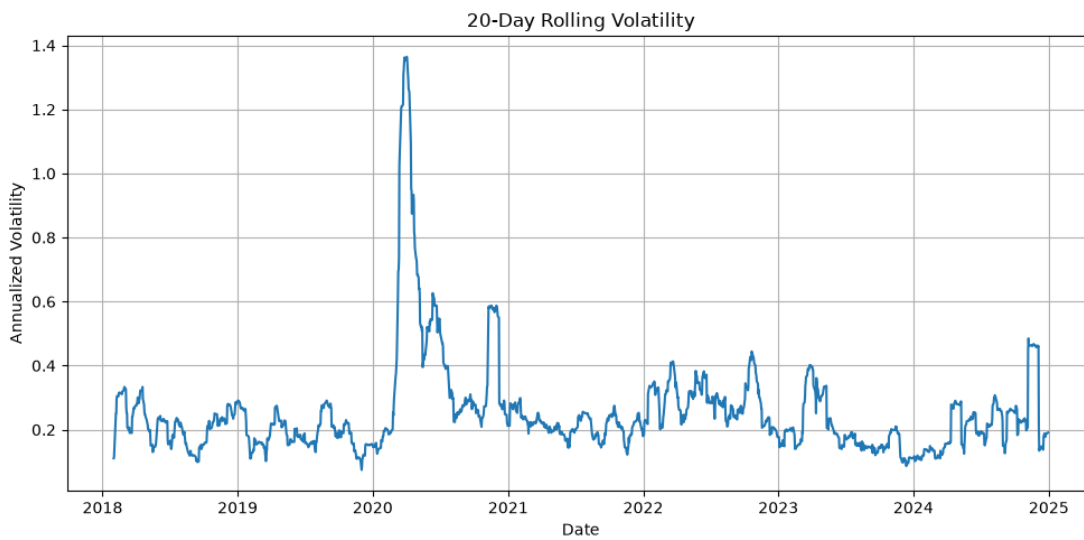
The 20-day and 50-day moving averages generated during feature engineering.

Figure 3. 20-Day Rolling Volatility



The annualized 20-day rolling historical volatility calculated from daily returns.

Figure 4. Distribution of Daily Returns



The distribution of daily returns after preprocessing. Most observations are concentrated around zero, while extreme returns occur less frequently.

5. Issues Encountered

Financial datasets collected from different sources required additional preprocessing

because of differences in formatting and trading calendars.

VIX is not a news-sentiment score. GitHub Actions run 34297832977 verified preprocessing, output validation, and artifact upload.

6. Plan for Week 3

Implement the Black–Scholes-based chooser option pricing model.

Configure the model parameters according to the project requirements.

Perform initial model validation through comparison with reference results.